CAPSTONE PROJECT REPORT

(Project Term January–May 2026)

ScropSec-IDS

Multi-Tenant Cloud-Native Intrusion Detection Platform with Scheduler-Driven Aggregation and LLM-Assisted Threat Triage

Submitted by

Chikati Yatesh Chandra Sai	Registration Number: 12319319
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Mendu Pavan Kalyan	Registration Number: 12302985

Project Group Number: 2RGC0079

Course Code: CSE439

Under the Guidance of

Chandan Kumar

School of Computer Science and Engineering

Lovely Professional University, Punjab

Academic Year: 2025–2026

TOPIC APPROVAL PERFORMA

School of Computer Science and Engineering

Program : P132-L::B.Tech. (Computer Science and Engineering) [Lateral Entry]

COURSE CODE : CSE439 **REGULAR/BACKLOG :** Regular **GROUP NUMBER :** 2RGC0079

Supervisor Name : Chandan Kumar **UID :** 33679 **Designation :** Assistant Professor

Qualification : ________________________ **Research Experience :** ________________________

SR.NO.	NAME OF STUDENT	Prov. Regd. No.	BATCH	SECTION
1	Chikati Yatesh Chandra Sai	12319319	2023	322FN
2	Konda Nagendar	12314014	2023	222LH
3	Mendu Pavan Kalyan	12302985	2023	322MR

SPECIALIZATION AREA : Network and Cyber Security **Supervisor Signature:** ____________________

PROPOSED TOPIC : ScropSec-IDS

Qualitative Assessment of Proposed Topic by PAC		
Sr.No.	**Parameter**	**Rating (out of 10)**
1	Project Novelty: Potential of the project to create new knowledge	7.00
2	Project Feasibility: Project can be timely carried out in-house with low-cost and available resources in the University by the students.	7.33
3	Project Academic Inputs: Project topic is relevant and makes extensive use of academic inputs in UG program and serves as a culminating effort for core study area of the degree program.	7.00
4	Project Supervision: Project supervisor’s is technically competent to guide students, resolve any issues, and impart necessary skills.	7.33
5	Social Applicability: Project work intends to solve a practical problem.	7.00
6	Future Scope: Project has potential to become basis of future research work, publication or patent.	7.33

PAC Committee Members		
PAC Member (HOD/Chairperson) Name: Dr. Atul Malhotra	UID: 18011	Recommended (Y/N): Yes
PAC Member (Allied) Name: Diksha Rani	UID: 28237	Recommended (Y/N): NA
PAC Member 3 Name: Sanjeev Kumar	UID: 18522	Recommended (Y/N): Yes

Final Topic Approved by PAC: **ScropSec-IDS**

Overall Remarks: Approved

PAC CHAIRPERSON Name: 16479::Dr. Parminder Singh **Approval Date:** 28 Apr 2026

DECLARATION

We hereby declare that the project work entitled ScropIDS – Multi-Tenant Cloud-Native Intrusion Detection Platform with Scheduler-Driven Aggregation and LLM-Assisted Threat Triage is an authentic record of our own work carried out as a requirement of the Capstone Project for the award of B.Tech degree in Computer Science / Data Science.from Lovely Professional University, under the guidance of Chandan Kumar during January to May 2026. All information furnished in this capstone project report is based on our own intensive work and is genuine. No part of this work has been submitted elsewhere for any other degree or diploma.

Project Group Number: 2RGC0079

Name of Student 1: Chikati Yatesh Chandra Sai Signature:

Date:

Name of Student 2: Konda Nagendar Signature:

Date:

Name of Student 3: Mendu Pavan Kalyan Signature:

Date:

CERTIFICATE

This is to certify that the declaration statement made by this group of students is correct to the best of my knowledge and belief. They have completed this Capstone Project under my guidance and supervision. The present work is the result of their original investigation, effort, and study. No part of the work has ever been submitted for any other degree at any university. The Capstone Project is fit for submission and partial fulfilment of the conditions for the award of B.Tech Degree in Computer Science / Data Science from Lovely Professional University, Punjab.

Signature and Name of the Mentor: Chandan Kumar
Designation: Professor
School of Computer Science and Engineering
Lovely Professional University, Punjab Date:

ACKNOWLEDGEMENT

The successful completion of this project would not have been possible without the invaluable guidance, support, and encouragement of numerous individuals. We take this opportunity to express our sincere gratitude to all those who contributed, directly or indirectly, to the realization of this work.

We are deeply grateful to our project supervisor, Chandan Kumar whose expert guidance, constructive criticism, and unwavering support throughout the development lifecycle were instrumental in shaping both the technical depth and academic quality of this project. Their mentorship extended beyond technical matters, instilling in us the values of rigorous inquiry and systematic problem solving.

We extend our heartfelt appreciation to the faculty members of the Department of Computer Science for their encouragement and for providing access to necessary resources and infrastructure. The broader academic environment at Lovely Professional University has been a constant source of intellectual stimulation.

We also acknowledge the open-source communities behind React, Node.js, MongoDB, Tailwind CSS, and Leaflet, whose exceptional tools and documentation formed the technical backbone of this platform. Additionally, we thank our peers and classmates who participated in user acceptance testing and provided candid feedback that significantly improved the user experience.

Finally, we owe a debt of gratitude to our families for their unconditional support, patience, and motivation throughout this academic journey.

ChikatiYatesh Chandra Sai

Konda Nagendar

Mendu Pavan Kalyan